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Two Transiting Candidates around Faint M Dwarfs from an Agent-Driven Search of *TESS* Full-Frame Images

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ABSTRACT

We report two transiting planet candidates around M dwarfs with *TESS* magnitudes 14.5–14.8, identified in an independent box-least-squares search of SPOC full-frame image (FFI) cutouts of 38,620 Sector 1 stars with $12 < T < 15$ and $R_\star < 0.6\,R_\odot$ that were neither 2-minute targets nor *TESS* Objects of Interest. Each signal is recovered independently in every available sector (3–9 sectors, 2018–2026), is located on the target in noise-normalized difference images, shows consistent odd and even depths and no secondary eclipse at any orbital phase, and is fit with a physical transit model. TIC 389723320 hosts a $4.6\,R_\oplus$ candidate at $P = 3.927167$ d and TIC 114949540 a $4.35\,R_\oplus$ candidate at $P = 2.739176$ d. A third signal, a 3.8% dip on TIC 259719966 at $P = 11.0772$ d, is an eccentric low-mass eclipsing binary: its secondary eclipse sits at phase 0.54, not 0.5, and is missed by a phase-0.5 test. Neither candidate appears in the TOI, Community Candidate, or confirmed-planet lists. The pipeline was built and operated with an AI coding agent under human direction; code and per-candidate vetting reports are public.

Keywords: Exoplanet detection methods (489) — Transit photometry (1709) — M dwarf stars (982) — Astronomy software (1855)

MOTIVATION

The *TESS* project's own FFI transit searches have concentrated on stars brighter than $T \approx 13.5$ (Huang et al. 2020; Kunimoto et al. 2022), leaving the fainter M-dwarf population, where a given planet produces the deepest transits, comparatively under-searched. We targeted that population directly: M dwarfs ($T_{\rm eff} < 4000$ K, $R_\star < 0.6\,R_\odot$) with $12 < T < 15$, excluding 2-minute targets, TOIs, Community Candidates, and known planet hosts. For Sector 1 this yields 40,328 unique stars, of which the 38,620 with contamination ratio below 1 were searched.

DATA AND SEARCH

For each star we extracted a 15×15 pixel cutout from the SPOC calibrated FFI sector cubes hosted on AWS (astrocut; Brasseur et al. 2019), defined a threshold aperture, summed the aperture flux, and removed shared systematics by regressing against the leading principal components of the non-aperture pixels (lightkurve; Lightkurve Collaboration et al. 2018). After normalization, a 33-hour Savitzky–Golay flatten, and a point-to-point-jump filter for pointing excursions, we ran a box least squares search (Kovács et al. 2002) over 0.3–13 d with durations 0.5–6 h. Candidates required S/N ≥ 9, depth ≥ 3000 ppm, at least two transits, duration no more than $1.5\times$ the maximum allowed by the TIC stellar density (Stassun et al. 2019), duty cycle below 15%, odd and even depths consistent within 3σ, and no secondary eclipse at phase 0.5 above 3σ. Twenty-two stars passed.

VETTING

Each flagged star was tested against Gaia DR3 (Gaia Collaboration et al. 2023) neighbors within 2 pixels, with the eclipse depth each neighbor would need to reproduce the signal; a flux-weighted centroid shift between in- and out-of-transit cadences; and a difference image (out-of-transit mean minus in-transit mean, per-frame plane-subtracted,

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divided by its per-pixel noise) whose peak locates the source of the dip (Bryson et al. 2013; Twicken et al. 2018). Of the 22, seven were located on a neighbor, one was a spotted rotator, one an unresolvable 1″ blend, and three showed anomalous centroid behavior. Every survivor was then searched independently in each sector in which *TESS* observed it (sectors from tess-point; Burke et al. 2020) and the sectors stitched to refine the period. Two high-S/N single-sector detections failed to repeat and were dropped. Three signals passed all tests; a final search for a secondary eclipse at every orbital phase (0.15–0.85, box of the transit duration, per-sector agreement required) removed one of them, described below.

CANDIDATES

Transit parameters were obtained with `batman` (Kreidberg 2015) and `emcee` (Foreman-Mackey et al. 2013), fitting epoch, period, $R_p/R_\star$, $a/R_\star$, impact parameter, and quadratic limb darkening with Gaussian priors (0.3 ± 0.1), supersampling each sector to its exposure time. Stellar parameters are from TIC v8. Figure 1 shows the phase-folded data and models.

TIC 389723320 ($T = 14.52$, $T_{\rm eff} = 3271$ K, $R_\star = 0.37\,R_\odot$, $d = 126$ pc). Six sectors (1, 28, 68, 95, 105, 106), 36 transits. $P = 3.927167 \pm 0.000001$ d, $T_0 = 2460899.3895 \pm 0.0005$ BJD, depth $12{,}850 \pm 690$ ppm, duration 1.50 ± 0.06 h, $b = 0.74$, $R_p = 4.62 \pm 0.13\,R_\oplus$, $T_{\rm eq} \approx 565$ K. Odd minus even depth 1100 ± 520 ppm (2.1σ); no secondary eclipse at any phase (< 620 ppm, 3σ). Difference images in sectors 68, 95, and 105 peak on the target at 14, 14, and 18σ. Transit-derived stellar density $4.1^{+1.5}_{-1.0}\,\rho_\odot$ versus 7.0 from the TIC, consistent within the b–$a/R_\star$ degeneracy.

TIC 114949540 ($T = 14.78$, $T_{\rm eff} = 3320$ K, $R_\star = 0.42\,R_\odot$, $d = 168$ pc). Three sectors (1, 27, 105), 25 transits. $P = 2.739176 \pm 0.000001$ d, $T_0 = 2461212.9505 \pm 0.0007$ BJD, depth 8930 ± 710 ppm, duration 1.55 ± 0.07 h, $b = 0.53$, $R_p = 4.35 \pm 0.17\,R_\oplus$, $T_{\rm eq} \approx 670$ K. Odd minus even -240 ± 450 ppm; no secondary eclipse at any phase (< 700 ppm, 3σ). Difference images in sectors 27 and 105 peak on the target at 15 and 8σ.

TIC 259719966 ($T = 13.80$, $T_{\rm eff} = 3443$ K, $R_\star = 0.49\,R_\odot$, $d = 147$ pc) showed a 3.8%, flat-bottomed, on-target dip at $P = 11.077205$ d in all nine sectors with equal odd and even depths and nothing at phase 0.5, and passed as a $0.94\,R_{\rm Jup}$ candidate. An all-phase search found a 0.92% secondary eclipse at phase 0.5405 in every sector (39σ; independently 22σ in MIT QLP light curves), implying $e\cos\omega = 0.061$, $e \approx 0.27$, and a surface-brightness ratio of 0.26, i.e. a ∼2700 K companion of $0.92\,R_{\rm Jup}$: a late-M dwarf or brown dwarf, not a planet. The eccentricity also resolves the circular-fit stellar density (5.9 vs 4.1 $\rho_\odot$). We retain it here as a cautionary example: a secondary-eclipse test confined to phase 0.5 is blind to eccentric binaries.

A residual search of the stitched light curves of both candidate systems, with the known transits masked, found no additional signal above ∼1000 ppm with three or more transits at $P < 30$ d.

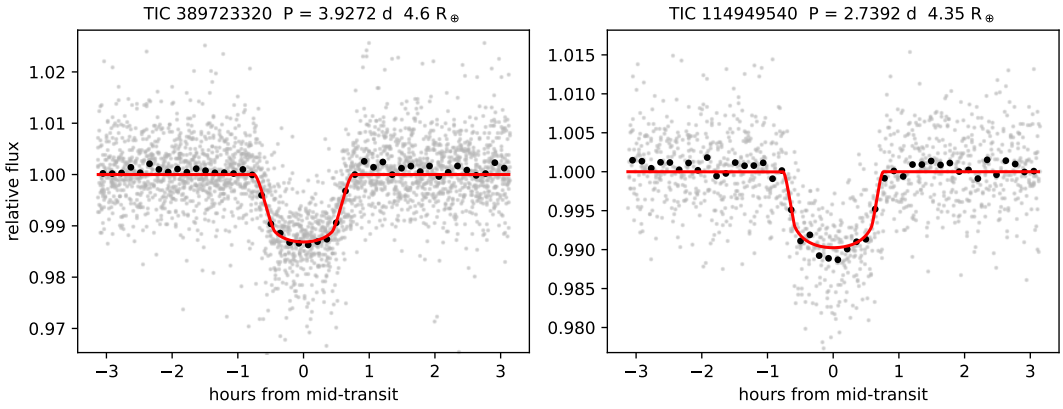


Figure 1. Phase-folded *TESS* FFI photometry (all sectors, gray) with the posterior-median transit model (red) for TIC 389723320 (left) and TIC 114949540 (right). Binned points in black.

CAVEATS AND AVAILABILITY

Both hosts are faint for radial-velocity follow-up. Neither candidate is a TOI, Community Candidate (ExoFOP list of 2026 October 24), or known planet host; none has an ADS literature match; and none lies within 30″ of an entry in

the DIAmante FFI candidate catalog (Montalto et al. 2020), the QLP faint-star TOI release (Kunimoto et al. 2022), the *TESS* eclipsing binary catalog (Prša et al. 2022), or the Gaia DR3 variability classification. The pipeline (target selection, parallel FFI sweep, vetting, multi-sector recovery, MCMC fitting, and report generation) was written and operated with Claude, an AI coding agent, directed by the author, who has no prior background in exoplanet science; we note this because the result bears on how such searches can be scaled. Code, pitfalls log, and per-candidate vetting reports are available at https://github.com/khariha/tess-hunt.

This research made use of data from the *TESS* mission, obtained from MAST at STScI. The *TESS* full-frame images used here can be found in MAST: 10.17909/t9-17nt-6c71. It made use of the NASA Exoplanet Archive 10.26133/NEA12 and ExoFOP 10.26134/ExoFOP5, operated by Caltech/IPAC under contract with NASA, and of data from the ESA mission Gaia, processed by the Gaia DPAC.

Software: astrocut (Brasseur et al. 2019), astropy, astroquery, batman (Kreidberg 2015), emcee (Foreman-Mackey et al. 2013), lightkurve (Lightkurve Collaboration et al. 2018), tess-point (Burke et al. 2020), Modal

REFERENCES

Brasseur, C. E., Phillip, C., Fleming, S. W., Mullally, S. E., & White, R. L. 2019, Astrocut: Tools for creating cutouts of TESS images, Astrophysics Source Code Library, ascl:1905.007

Bryson, S. T., Jenkins, J. M., Gilliland, R. L., et al. 2013, PASP, 125, 889

Burke, C. J., Levine, A., Fausnaugh, M., et al. 2020, TESS-Point: High precision TESS pointing tool, Astrophysics Source Code Library, ascl:2003.001

Foreman-Mackey, D., Hogg, D. W., Lang, D., & Goodman, J. 2013, PASP, 125, 306

Gaia Collaboration, Vallenari, A., Brown, A. G. A., et al. 2023, A&A, 674, A1

Huang, C. X., Vanderburg, A., Pál, A., et al. 2020, RNAAS, 4, 204

Kovács, G., Zucker, S., & Mazeh, T. 2002, A&A, 391, 369

Kreidberg, L. 2015, PASP, 127, 1161

Kunimoto, M., Daylan, T., Guerrero, N., et al. 2022, ApJS, 259, 33

Lightkurve Collaboration, Cardoso, J. V. d. M., Hedges, C., et al. 2018, Lightkurve: Kepler and TESS time series analysis in Python, Astrophysics Source Code Library, ascl:1812.013

Montalto, M., Borsato, L., Granata, V., et al. 2020, MNRAS, 498, 1726

Prša, A., Kochoska, A., Conroy, K. E., et al. 2022, ApJS, 258, 16

Stassun, K. G., Oelkers, R. J., Paegert, M., et al. 2019, AJ, 158, 138

Twicken, J. D., Catanzarite, J. H., Clarke, B. D., et al. 2018, PASP, 130, 064502